

Theories of Straight beams: Part 1

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1 Vocabulary

Young's Modulus Denoted by E , Young's Modulus quantifies the relationship between stress and strain and is represented by the equation:

$$E = \frac{\sigma}{\epsilon} \quad (1)$$

Young's Modulus quantifies the material's ability to resist axial elastic deformation. Figure 1 represents the stress-strain curve for steel 1018 under various conditions. The Young's Modulus is given by finding the slope of the portion where the slope is still linear and applies only while in this region. After plastic deformation has occurred, this relationship no longer holds. Generally, ceramics have high E values, metals have intermediate E values, and polymers have low E values.

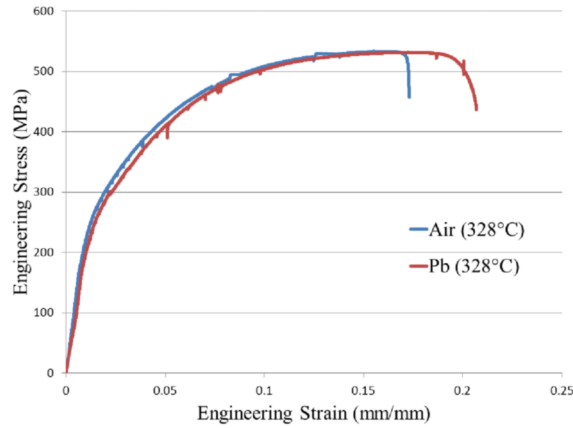


Figure 1: Engineering stress-strain curves for 1018 steel in air and molten lead both at 328°C showing similar tensile behavior.²

Poisson's Ratio Denoted by ν , Poisson's Ratio is a ratio of the transverse strain and the axial strain that a specimen experiences while experiencing an applied force(s). This ratio describes how much the specimen will expand or contract (in a direction perpendicular to the direction of the applied force). It is defined by

$$\nu = -\frac{\epsilon_{transverse}}{\epsilon_{axial}} \quad (2)$$

where $\epsilon_{transverse}$ is the strain perpendicular to the applied load and ϵ_{axial} is the strain parallel to the applied load. Its negative to ensure positive values of ν ($\epsilon_{transverse}$ should always have the opposite sign as ϵ_{axial}).

²Engineering stress-strain curves for 1018 steel in air and molten lead, both at 328°C [Figure]. ResearchGate. https://www.researchgate.net/figure/Engineering-stress-strain-curves-for-1018-steel-in-air-and-molten-lead-both-at-328C_fig100277197915

Shear Modulus Similar to Young's modulus of elasticity, the shear modulus is a constant that quantifies how resistant a material is to deformation while experiencing shear loading. Denoted by G , its represented by the equation:

$$G = \frac{\tau}{\gamma} \quad (3)$$

where τ is the shear stress and γ is the shear strain. It is connected with the Young's modulus and Poisson's ratio through the equation:

$$E = 2G(1 + \nu) \quad (4)$$

or if you rearrange to solve for G ,

$$G = \frac{E}{2(1 + \nu)} \quad (5)$$

Similar to the Young's modulus, the shear modulus remains constant until plastic deformation occurs, at which point G will vary. Generally, ceramics have high G values, metals have intermediate G values, and polymers have low G values.

Standard Beam Coordinate System The standard beam coordinate system is a coordinate system that has been defined across industry to standardize the nomenclature when analyzing beams under loading. In this, the x-axis is always defined as running along the length of the beam. The y-axis is the horizontal axis along the cross-sectional view, located in the center of the beam. Likewise, the z-axis is the vertical axis along the cross-sectional view, located at the center of the beam. A moment about the y-axis will cause the beam to move up or down, and a moment about the z-axis will cause the beam to move right or left. A force applied along the x-axis is defined as the axial normal force. A force applied parallel to the y-axis or the z-axis generates shear. A moment applied about the x-axis is considered torsion, and moments about either the y-axis or the z-axis are both bending moments. This system is based around a right-handed coordinate system. See figure 2 for an illustration of this system. The origin of this coordinate system is centered at one of the beam's nodes.

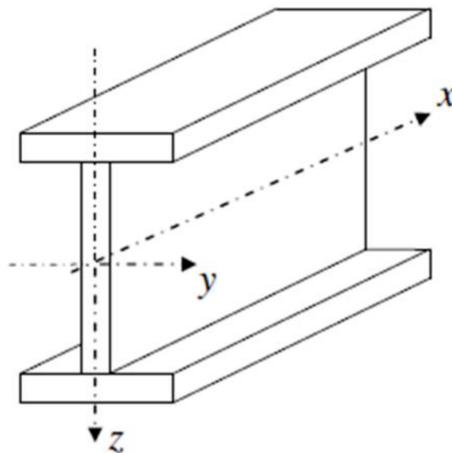


Figure 2: The standard beam coordinate system.³

³Abuteir, 2021. Orientation of beam coordinates [Figure].
https://www.researchgate.net/figure/Orientation-of-beam-coordinates_fig4354462528

Cross-Sectional Moment of Inertia (About Different Axis) Required to calculate the stresses in a beam and the deflections caused by these stresses, the moment of inertia (I) is a measure of how the beam's cross-sectional area is distributed relative to a specific axis. If you find I_x , you get the moment of inertia relative to the x-axis. Similarly, I_y is relative to the y-axis and I_z is relative to the z-axis. Generally, only I_x and I_y are used in beam theory when finding the beam's deflection, and I_z is used to find the beam's twist (torsional deflection). I is generally given by the equation:

$$I = \int_A r^2 dA \quad (6)$$

however, generally it is already calculated and expressed similar to:

$$I_x = \frac{bh^3}{12} \quad (7)$$

The above equation is the moment of inertia relative to the x-axis for a beam with a rectangular cross-section. This equation will vary depending on the cross-sectional shape, and the equations for varying shapes can be found online. The moment of inertia is critical to calculating stresses in a beam.

Cross-Sectional Polar Moment of Inertia The polar moment of inertia measures a beam's cross-sectional ability to resist twisting about an axis perpendicular to the cross-section. Denoted by J , the polar moment of inertia about the x-axis is given by the equation:

$$J_x = I_y + I_z \quad (8)$$

The polar moment of inertias about the y-axis and z-axis are given respectively by equations 9 and 10.

$$J_y = I_x + I_z \quad (9)$$

$$J_z = I_x + I_y \quad (10)$$

If needed, you can also find these by integrating.

$$J_{polar,x} = \int_A (y^2 + z^2) dA \quad (11)$$

You can use the same relationship defined in equations 8-10 to find $J_{polar,y}$ and $J_{polar,z}$. If x, y or z are on the left side of the equation, the other two variables (x, y, or z) should be on the other side. Explicitly, the equations are:

$$J_{polar,y} = \int_A (z^2 + x^2) dA \quad (12)$$

$$J_{polar,z} = \int_A (y^2 + x^2) dA \quad (13)$$

For help finding the moments of inertia, reference *Cross-Sectional Moment of Inertia (About Different Axis)* from this document.

2 Theory

Euler-Bernoulli Beam Theory Assumptions The Bernoulli–Euler beam theory is based on certain simplifying assumptions, known as the Bernoulli–Euler hypothesis, concerning the kinematics of bending deformation. The hypothesis states that straight lines perpendicular to the beam axis before deformation remain (a) straight, (b) inextensible, and (c) perpendicular to the tangent line to the beam axis after deformation⁴. In essence, the assumption is that any given cross-section throughout the beam will not warp or distort in any way during bending. They may rotate, but they will stay orthogonal to the beam’s centerline. All geometry changes due to deflection are considered negligible.

It assumes that the beam is linearly elastic and that no plastic deformation has occurred during bending; there must be no creep, plasticity, or other nonlinear deformation. It assumes that the primary source of stress is axial stress, not shear stress, and the shear stresses must be minimal). For the Euler-Bernoulli theory to work, all beams must have a high aspect ratio.

Relationship Between Euler-Bernoulli Displacement Fields and Infinitesimal Strains

The displacement field is a vector field that assigns to every point within a beam the change in its position due to deformation. The Bernoulli–Euler hypothesis leads to the following displacement field:

$$u_1(x, y, z) = u(x) - z \frac{dw}{dx}, u_2 = 0, u_3(x, y, z) = w(x) \quad (14)$$

where (u, v, w) are the displacements of a point on the x-axis in the $x_1 = x$, $x_2 = y$, $x_3 = z$ coordinate directions, respectively⁵ (See Figure 3).

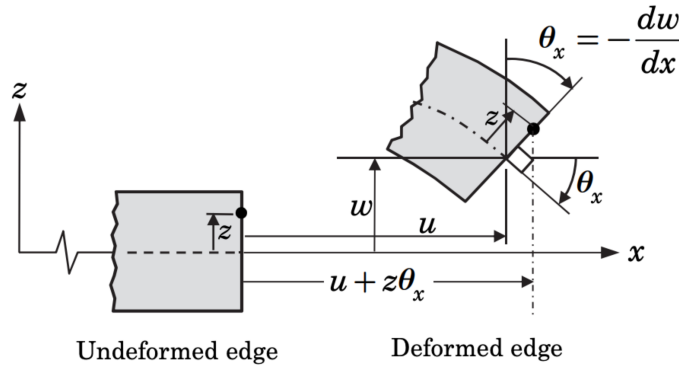


Figure 3: The standard beam coordinate system.⁶

Infinitesimal strains describe how a beam deforms when displacements and rotations are very small compared to the beam’s dimensions. The infinitesimal strains are given by the

⁴JN Reddy, Energy principles and Variational Methods in Applied Mechanics, 2017. pg. 3

⁵JN Reddy, Energy principles and Variational Methods in Applied Mechanics, 2017. pg. 3

⁶JN Reddy, Energy principles and Variational Methods in Applied Mechanics, 2017. pg. 3

equation:

$$\epsilon_{xx} = \frac{\partial u_1}{\partial x_1} = \frac{du}{dx} - z \frac{d^2 w}{dx^2}, 2\epsilon_{xx} = \frac{\partial u_1}{\partial x_3} + \frac{\partial u_3}{\partial x_1} = -\frac{dw}{dx} + \frac{dw}{dx} = 0 \quad (15)$$

All other strains identically simplify to zero. Note that you must find the displacement field to calculate the infinitesimal strains. This expression indicates that axial stretching of the neutral axis is governed by the derivative of the displacement ($u'_0(x)$), while bending introduces a linear variation of strain through the thickness proportional to the curvature $w''(x)$.

Infinitesimal Strain's Relationship with Engineering Strain Engineering strain measures how much a beam stretches or shortens relative to its original length, and is given by the equation:

$$\epsilon = \frac{\Delta L}{L_0} \quad (16)$$

which is identical to:

$$\epsilon = \frac{du}{dx} \quad (17)$$

For a description of Infinitesimal strains, see *Relationship Between Euler-Bernoulli Displacement Fields and Infinitesimal Strains* on page 5 of this document. Under linear conditions, engineering strain becomes identical to the infinitesimal strain

$$\epsilon_{xx} = \frac{\partial u}{\partial x} \quad (18)$$

Engineering strain is the 1D representation of the infinitesimal strain. Under nonlinear conditions or large displacements, this no longer holds true.

Euler-Bernoulli Force and Moment Equilibrium Equations, Shear and Axial Stress

This section aims to answer the questions 1) How does the Euler-Bernoulli force and moment equilibrium equations lead to a derivation of the axial and shear stress and 2) Why in this theory do we need another equation to find the shear stress?

Summing the forces in the x- and z- directions as well as the moments at the right end of a beam gives the following equilibrium equations:

$$\sum F_x = 0 : -\frac{dN}{dx} = f(x) \quad (19)$$

$$\sum F_z = 0 : -\frac{dV}{dx} = q(x) \quad (20)$$

$$\sum M_y = 0 : V - \frac{dM}{dx} = 0 \quad (21)$$

where $N(x)$ is the net axial force, $M(x)$ is the net bending moment about the y-axis, and $V(x)$ is the net transverse shear force (see Figure 4) on the beam's cross section:

$$N(x) = \int_A \sigma_{xx} dA, M(x) = \int_A \sigma_{xx} z dA, V(x) = \int_A \sigma_{xz} dA, \quad (22)$$

where dA denotes an area element of the cross section ($dA = dydz$). The set (N, M, V) are known as the stress resultants (because they result from the stresses in the beam)⁷.

⁷JN Reddy, Energy principles and Variational Methods in Applied Mechanics, 2017. pg. 4

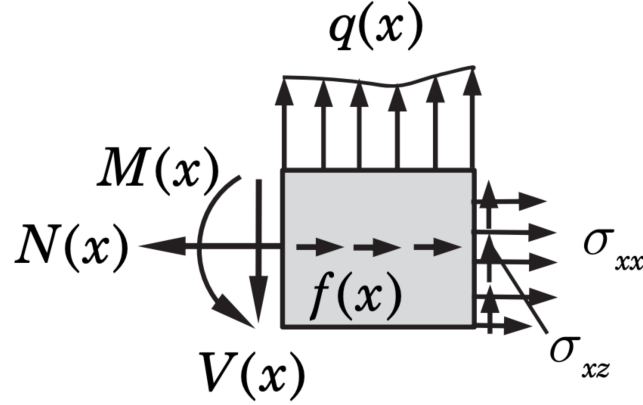


Figure 4: The standard beam coordinate system.⁸

The stresses in the beam can be computed using the linear elastic constitutive relation for an isotropic but possibly inhomogeneous material (i.e., the material properties may be functions of x and z):

$$\sigma_{xx}(x, z) = E(x, z)\epsilon_{xx}(x, z) = E\left(\frac{du}{dx} - z\frac{d^2w}{dx^2}\right) \quad (23)$$

$$\sigma_{xz}(x, z) = 2G(x, z)\epsilon_{xz}(x, z) = 0 \quad (24)$$

Although the transverse shear stress computed using constitutive equation is zero (hence, $V = 0$), it cannot be zero in a beam because the force equilibrium, Eq. (20) is violated. This is the inconsistency resulting from the Bernoulli–Euler beam hypothesis. Therefore, to respect the physical requirement of $V \neq 0$, we do not compute the shear stress from $\sigma_{xz} = 2G\epsilon_{xz}$; it is computed using the shear force V obtained from the equilibrium condition in Eq. (21), namely, $V = dM/dx$. For more information on this derivation, see pages 3-6 of *JN Reddy, Energy principles and Variational Methods in Applied Mechanics, 2017.*

Differences between the Euler-Bernoulli and the Timoshenko theories’ assumptions There are several differences in the assumptions of the Euler-Bernoulli theory and the Timoshenko theory. The Timoshenko beam theory is based on a relaxation of Bernoulli’s normality conditions, specifically that straight lines perpendicular to the beam axis before deformation remain perpendicular to the tangent line to the beam axis after deformation. In other words, the transverse normal has rotation ϕ_x that is not the same as the slope dw/dx . The difference between these two quantities is the transverse shear strain⁹. The Timoshenko theory includes shear deformation, while the Euler-Bernoulli Equation does not. Shear strain in this case is considered linear throughout the beam’s cross-section; however, a shear correction factor k is used to account for the real non-uniform distribution (See *Shear Correction Factor k* on pg. 8 for more information).

The relaxed Bernoulli–Euler hypothesis leads to the following displacement field for the Timoshenko beam theory (see Figure 5)

⁸JN Reddy, Energy principles and Variational Methods in Applied Mechanics, 2017. pg. 3

⁹JN Reddy, Energy principles and Variational Methods in Applied Mechanics, 2017. pg. 7

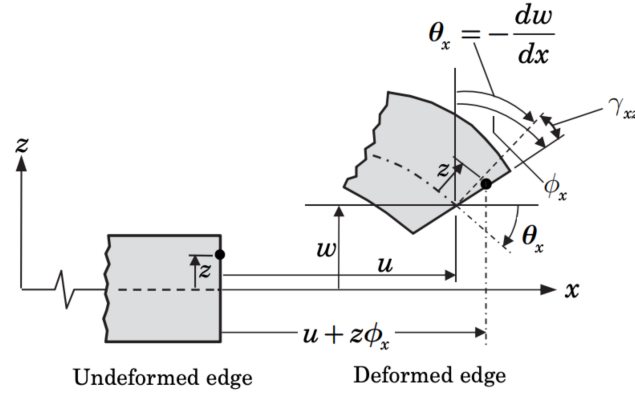


Figure 5: Kinematics of deformation in the Timoshenko beam theory.¹⁰

$$u_1(x, y, z) = u(x) + z\phi_x, u_2 = 0, u_3(x, y, z) = w(x) \quad (25)$$

where ϕ_x is the rotation of a transverse normal line¹¹. The Euler-Bernoulli equation has an input of one degree of freedom ($w(x)$); however, the Timoshenko theory has two degrees of freedom ($w(x), \theta(x)$). The Euler-Bernoulli equation is generally better for long, slender beams while the Timoshenko theory allows for more accurate analysis of short, stubby beams.

Shear Correction Factor k The Timoshenko beam theory assumes that shear deformation is evenly distributed throughout the height of a beam's cross-section. This assumption allows for the shear deformation to be calculated using the equation:

$$\gamma = \theta(x) - w'(x) \quad (26)$$

In reality, the shear distribution varies throughout any given cross-section, being the highest at the top and bottom of the beam, while being zero in the middle. If the top of the beam is in compression, the bottom will be in tension; if the bottom is in compression, the top will be in tension. Unlike the Euler-Bernoulli theory, the Timoshenko theory allows for these shear strains to be included; however, it inaccurately models how the strain is distributed. Modeling the strain as uniform rather than a parabolic curve allows for simplification of the math and is generally accurate when using a shear correction factor k . k is used to account for any incongruities that arise between a real shear strain distribution and that calculated using the Timoshenko beam theory. k will be different for different cross-sectional shapes. For a rectangular cross-section, its generally $\frac{5}{6}$, for a solid circle $\frac{9}{10}$, for a I-beam 1 (in the web), and so forth. You can generally use tabulated values to find k for any given cross-sectional shape.

¹⁰JN Reddy, Energy principles and Variational Methods in Applied Mechanics, 2017. pg. 7

¹¹JN Reddy, Energy principles and Variational Methods in Applied Mechanics, 2017. pg. 7

2.1 Force/Moment/Stress Equations and Their Differences Between Theories

The following equations are all under Euler-Bernoulli beam theory assumptions. The equations for forces, moments, and shear are:

$$\sum F_x = 0 : -\frac{dN}{dx} = f(x) \quad (27)$$

$$\sum F_z = 0 : -\frac{dV}{dx} = q(x) \quad (28)$$

$$\sum M_y = 0 : V - \frac{dM}{dx} = 0 \quad (29)$$

where $N(x)$ is the net axial force, $M(x)$ is the net bending moment about the y-axis, and $V(x)$ is the net transverse shear force on the beam's cross section¹²:

$$N(x) = \int_A \sigma_{xx} dA, M(x) = \int_A \sigma_{xx} z dA, V(x) = \int_A \sigma_{xz} dA \quad (30)$$

where dA denotes an area element of the cross section ($dA = dydz$). The stresses in the beam for an isotropic but possibly inhomogeneous material can be calculated using¹³:

$$\sigma_{xx}(x, z) = E(x, z)\epsilon_{xx}(x, z) = E\left(\frac{du}{dx} - z\frac{d^2u}{dx^2}\right) \quad (31)$$

$$\sigma_{xz}(x, z) = 2G(x, z)\epsilon_{xz}(x, z) = 0 \quad (32)$$

The stress resultants $N(x)$ and $M(x)$ can be related to the displacements u and w and the stress σ_{xx} using:

$$N(x) = \int_A \sigma_{xx} dA = A_{xx} \frac{du}{dx} + B_{xx} \left(\frac{d^2u}{dx^2}\right) \quad (33)$$

$$M(x) = \int_A z \sigma_{xx} dA = B_{xx} \frac{du}{dx} + D_{xx} \left(-\frac{d^2u}{dx^2}\right) \quad (34)$$

where A_{xx} is the axial stiffness, B_{xx} is the bending-stretching stiffness, and D_{xx} is the bending stiffness¹⁴.

The following equations are under the assumptions of the Timoshenko beam theory.

The stresses in a beam are:

$$\sigma_{xx}(x, z) = E(x, z)\epsilon_{xx}(x, z) = E\left(\frac{du}{dx} + z\frac{d\phi_x}{dx}\right) \quad (35)$$

$$\sigma_{xz}(x, z) = 2G(x, z)\epsilon_{xz}(x, z) = G(x, z)\left(\phi_x + \frac{dw}{dx}\right) \quad (36)$$

¹²JN Reddy, Energy principles and Variational Methods in Applied Mechanics, 2017. pg. 4

¹³JN Reddy, Energy principles and Variational Methods in Applied Mechanics, 2017. pg. 4

¹⁴JN Reddy, Energy principles and Variational Methods in Applied Mechanics, 2017. pg. 4

Note that the shear strain (hence, shear force) is not zero, removing the inconsistency of the Bernoulli–Euler beam theory¹⁵. The stress resultants (N, M) in the Timoshenko beam theory have the form:

$$N(x) = \int_A \sigma_{xx} dA = A_{xx} \frac{du}{dx} + B_{xx} \left(\frac{d\phi_x}{dx} \right) \quad (37)$$

$$M(x) = \int_A z \sigma_{xx} dA = B_{xx} \frac{du}{dx} + D_{xx} \left(\frac{d\phi_x}{dx} \right) \quad (38)$$

$$V(x) = \int_A K_s \sigma_{xz} dA = K_s S_{xz} \left(\phi_x + \frac{dw}{dx} \right) \quad (39)$$

where K_s is the shear correction coefficient introduced to correct the energy loss due to the constant state of transverse shear stress σ_{xz} and S_{xz} is the shear stiffness, given by:

$$S_{xz} = \int_A G(x, z) dA \quad (40)$$

Analysis

As noted previously, the main difference between these equations is that shear deformation is neglected in the Euler-Bernoulli equations and included in the Timoshenko equations. For example, when finding the stress resultants using the Timoshenko beam theory, equation 39 (also shown below in Eq. 41) is used, which is not valid for the Euler-Bernoulli theory. Note that the shear correction factor K_s is used in Eq. 39 to account for the inconsistency in how shear is modeled in the Timoshenko theory, whereas equation 30 (Shown below as Eq. 42) omits the K_s terms:

$$V(x) = \int_A K_s \sigma_{xz} dA = K_s S_{xz} \left(\phi_x + \frac{dw}{dx} \right) \quad (41)$$

$$V(x) = \int_A \sigma_{xz} dA \quad (42)$$

Due to the Timoshenko theory's inclusion of shear, equation 32 from the Euler-Bernoulli theory can no longer be set equal to zero. Equation 43 is under the Euler-Bernoulli theory, and equation 44 is under the Timoshenko theory. Both are shown here for convenience.

$$\sigma_{xz}(x, z) = 2G(x, z)\epsilon_{xz}(x, z) = G(x, z)\left(\phi_x + \frac{dw}{dx}\right) \quad (43)$$

$$\sigma_{xz}(x, z) = 2G(x, z)\epsilon_{xz}(x, z) = G(x, z)\left(\phi_x + \frac{dw}{dx}\right) \quad (44)$$

The stress resultant equations under the Timoshenko theory also expand upon those under the Euler-Bernoulli theory (there are two equations that allow you to solve for each one of the three stress resultants), allowing you to more easily find these stress resultants (See Eq.'s 30, 37-39).

¹⁵JN Reddy, Energy principles and Variational Methods in Applied Mechanics, 2017. pg. 4

3 Exploration

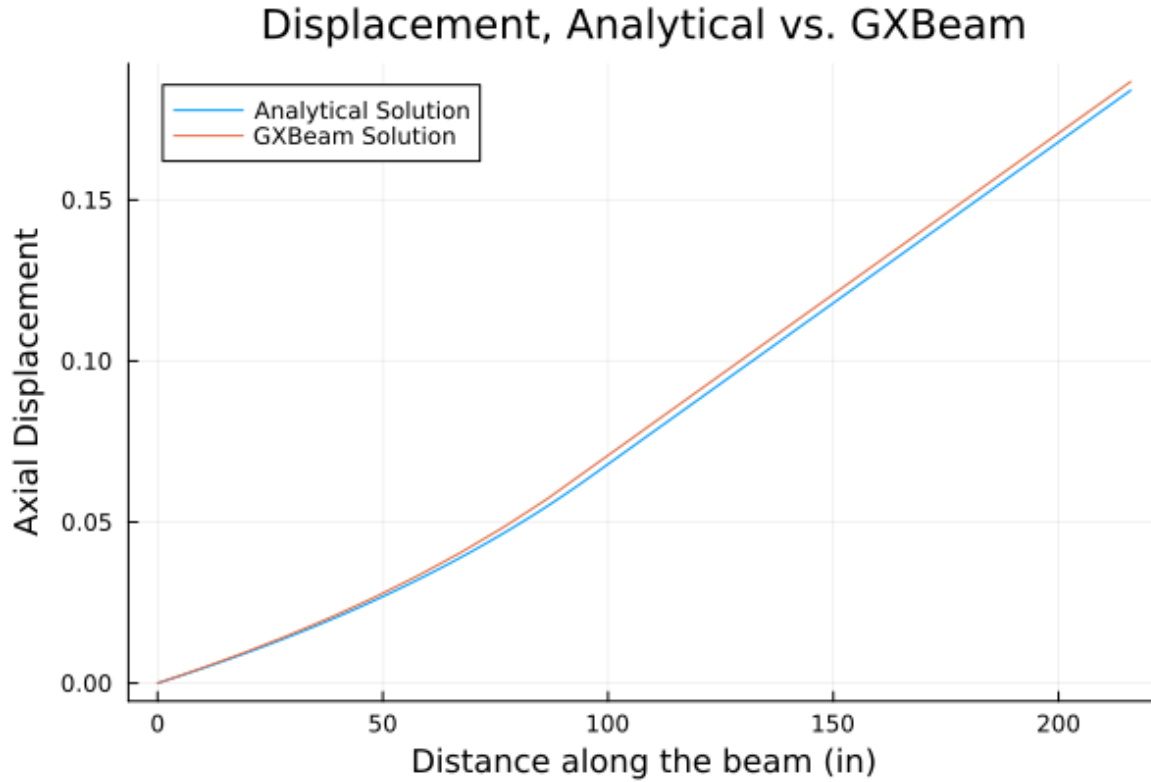


Figure 6: Comparison of the axial displacement of a beam using an analytical solution and GXBeam¹⁷

¹⁷The beam used in this example is the the example beam A2.1 from <https://mechanics.tamu.edu/wp-content/uploads/2017/03/Appendix-A_{Exact} – Analytical – Solutions.pdf>

¹⁸The beam used in this example is the the example beam A3.1 from <https://mechanics.tamu.edu/wp-content/uploads/2017/03/Appendix-A_{Exact} – Analytical – Solutions.pdf>

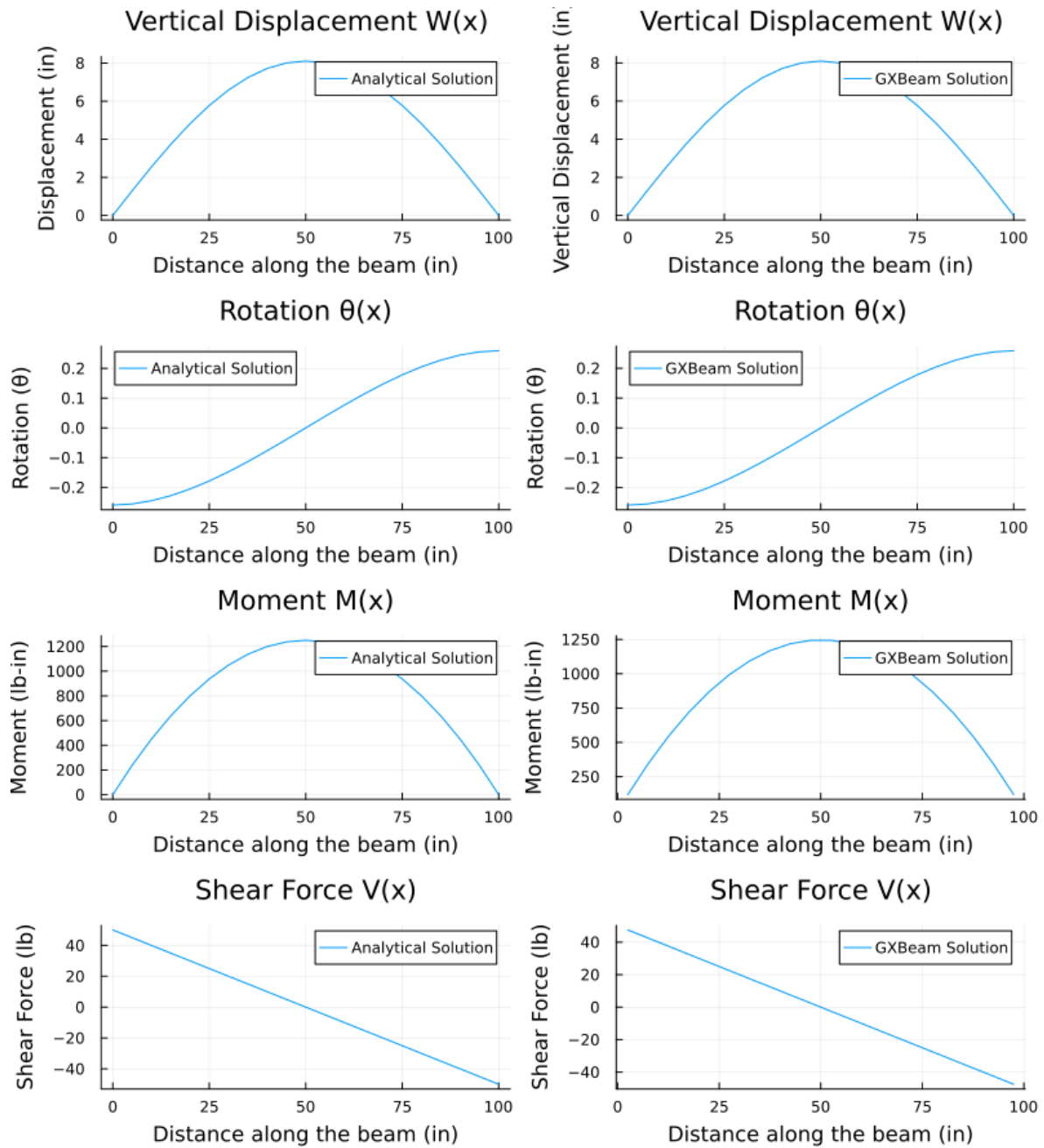


Figure 7: Comparison of results from the analytical solution vs GXBeam¹⁸